

Mapping Access Under Occupation

An accessibility analysis of the Budapest ghetto, 1944–45

This PDF is a print of the project website: atlas plates from the interactive map, the completed-research one-pager, the method note, working paper, CAGR briefing, teaching module, outreach letters, and data provenance. It is one city, eight months, not a GIS of “the Holocaust.”

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1. Atlas plates — occupation, yellow-star houses, sealed ghetto, June vs sealed
2. One-pager — completed research brief
3. Method — occupation-adapted LUPTAI
4. Working paper
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7. Outreach emails
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INTERACTIVE CASE STUDY

How spatial control changed daily survival

A LUPTAI-style accessibility index, adapted for occupation, over the Budapest Jewish population from the yellow-star houses of June 1944 to the sealed Pest ghetto of January 1945. Click a gold house or a coloured resource on the map.

APR-JUN 15

Occupation, still at home

24 JUN

Yellow-star houses

15 OCT

Arrow Cross coup

29 NOV

Two ghettos, still unsealed

10 DEC

Pest ghetto sealed

10 JAN

Two gates locked

After 19 March the yellow star, shopping-hour limits, and bans on baths, hotels, and cinemas shrink the day without yet redrawing the map of home. Jews still live across the city. Access is already a temporal problem.

35.2

MEAN COMPOSITE

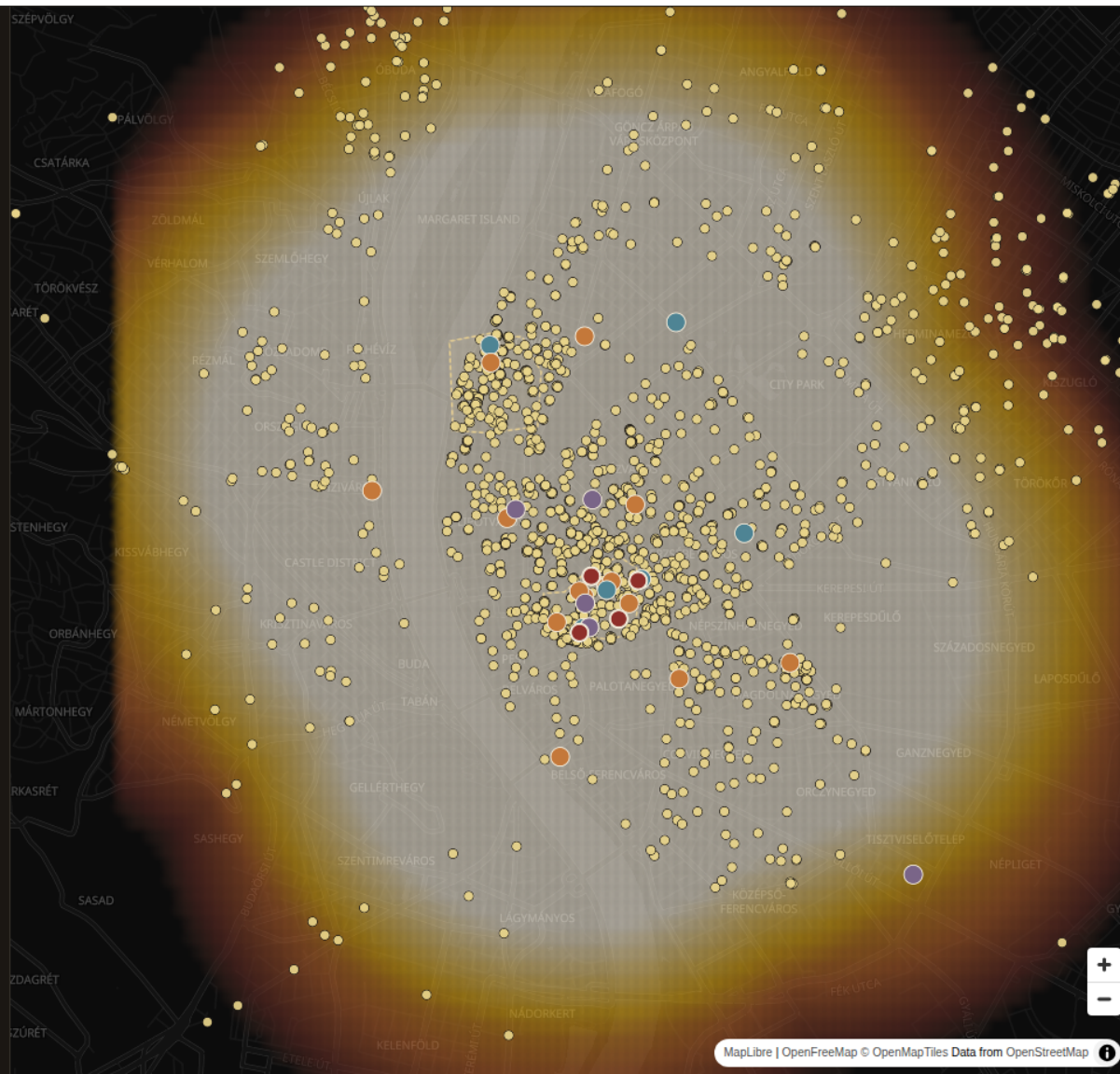
1,952

RESIDENCES IN
MODEL

10h

LEAVE-HOME WINDOW

92.2%

DROP, JUNE –
SEALED

INDEX

What is being measured

LUPTAI bands walking time to destinations, then blends purposes. Here the purposes are food, medical care, and work. Occupation adds four modifiers LUPTAI never needed: legal reach, the leave-home window, destination capacity, and falling walk speed.

Purpose

COMPOSITE

FOOD

MEDICAL

WORK

None / sealed Higher access

Layers

RESIDENCES

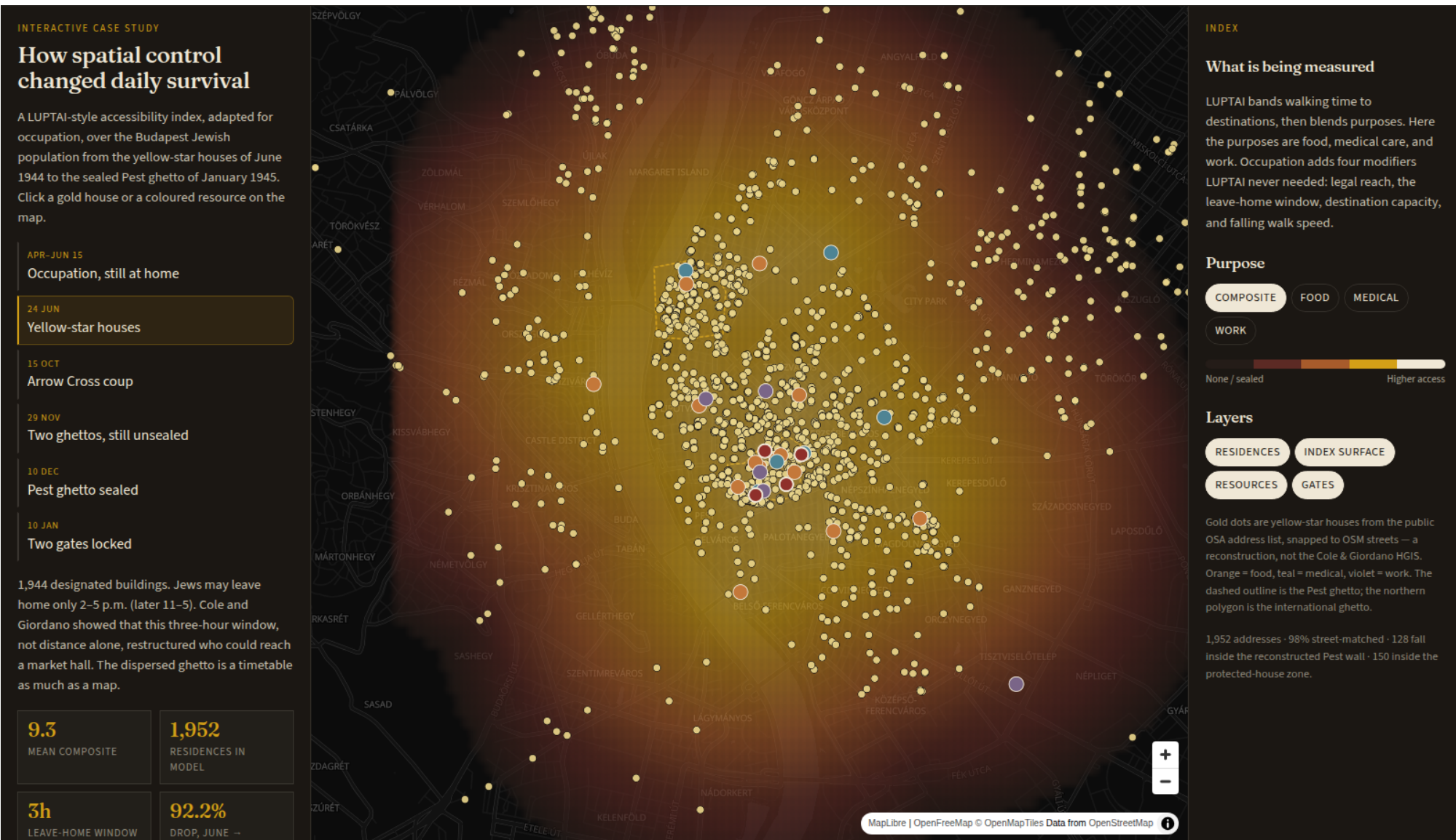
INDEX SURFACE

RESOURCES

GATES

Gold dots are yellow-star houses from the public OSA address list, snapped to OSM streets — a reconstruction, not the Cole & Giordano HGIS. Orange = food, teal = medical, violet = work. The dashed outline is the Pest ghetto; the northern polygon is the international ghetto.

1,952 addresses · 98% street-matched · 128 fall inside the reconstructed Pest wall · 150 inside the protected-house zone.



1,944 designated buildings. Leave-home window: three hours. The map of homes has changed; the map of destinations has not.

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Pest ghetto sealed

10 JAN

Two gates locked

Four gates, a fence the Jewish Council had to pay for, no individual cooking. City markets and both central hospitals sit outside the wall. A kitchen 200 metres away scores 'high' on a naïve distance index and still delivers 781 kcal. That gap is the point of the method.

0.7

MEAN COMPOSITE

278

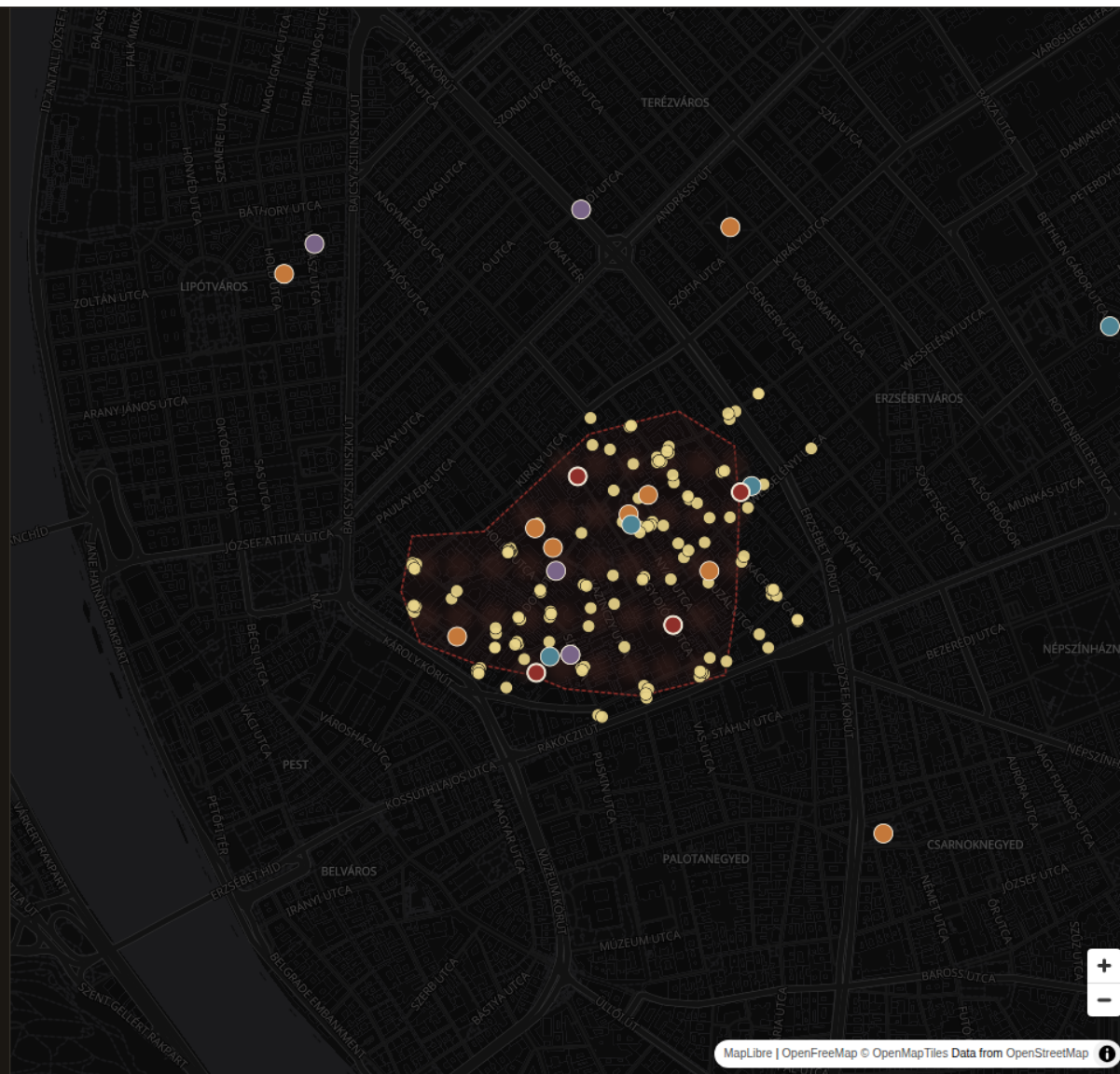
RESIDENCES IN MODEL

0.4h

LEAVE-HOME WINDOW

92.2%

DROP, JUNE -



INDEX

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INDEX SURFACE

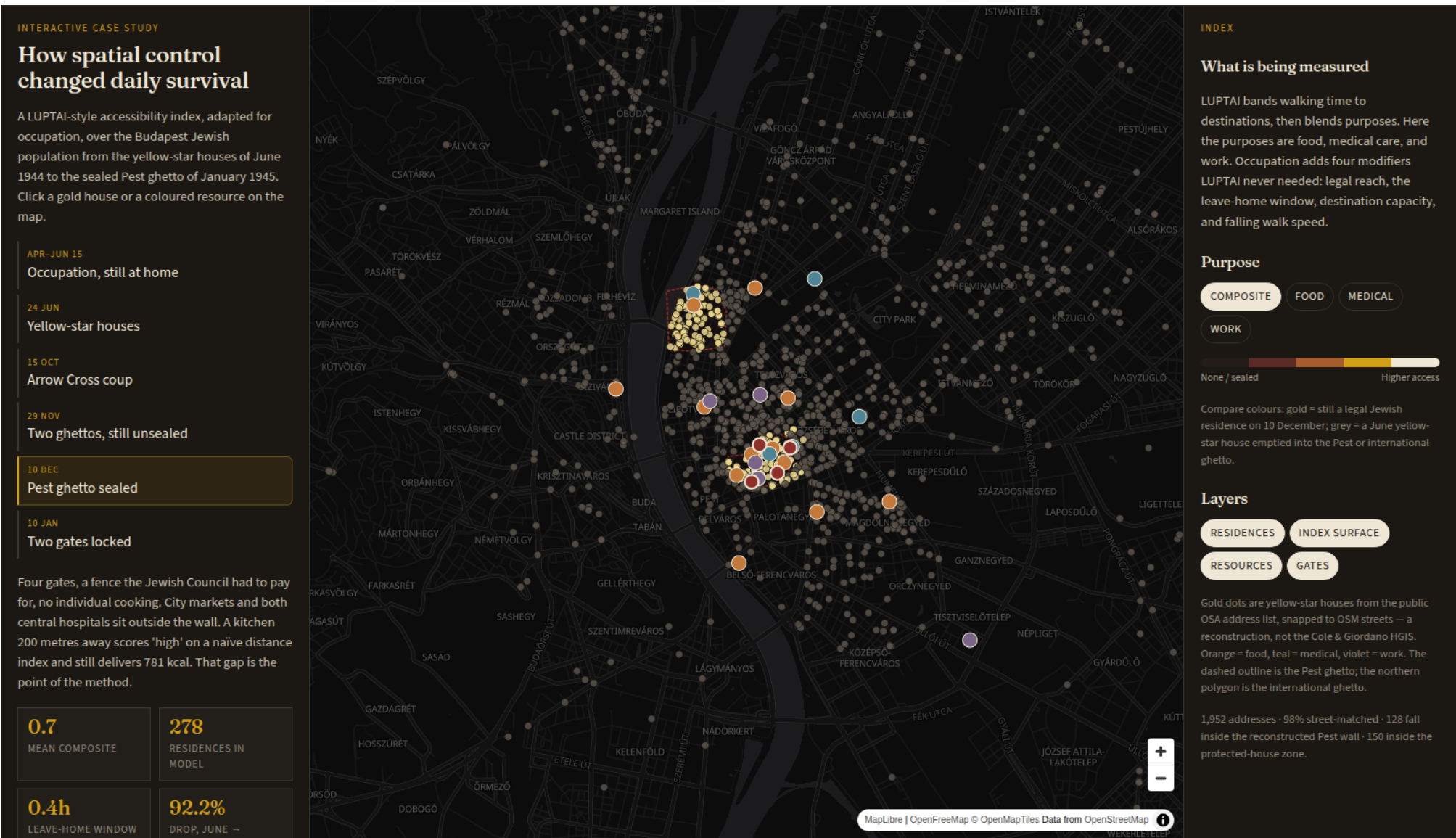
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1,952 addresses · 98% street-matched · 128 fall inside the reconstructed Pest wall · 150 inside the protected-house zone.

MapLibre | OpenFreeMap © OpenMapTiles Data from OpenStreetMap



Gold houses are still legal Jewish residences on 10 December. Grey houses are yellow-star buildings emptied into the two ghettos.



Mapping Access Under Occupation

An accessibility analysis of the Budapest ghetto, 1944–45

Spatial control in Budapest was a timetable before it was a wall. Layering a LUPTAI-style accessibility index onto the Cole–Giordano historical GIS shows that Jewish residents lost most of their practical access to food, medical care, and work in June 1944 — when they could leave home only three hours a day — and that sealing the Pest ghetto in December inverted the map: walking-time bands to kitchens went up while calories, hospitals, and legal reach collapsed.

73%

DROP IN MEAN COMPOSITE ACCESS, OCCUPATION → 24 JUNE, BEFORE ANY WALL

91%

FURTHER DROP AMONG REMAINING RESIDENCES, JUNE → SEALED GHETTO, 10 DEC

79%

OF 1,944 YELLOW-STAR HOUSES EMPTIED INTO THE TWO GHETTOS OR OFF THE LEGAL MAP

781

KCAL/ADULT/DAY INSIDE THE WALL, AGAINST A 2,200 KCAL NOTE ON THE SAME COUNCIL SHEET

THE CASE

One city, eight months, not “the Holocaust.” German occupation of Hungary on 19 March 1944; yellow-star houses from 24 June; Arrow Cross coup on 15 October; Pest ghetto and international ghetto from late November; wall sealed 10 December; two gates locked 10 January 1945; liberation 17–18 January. The empirical floor is the Holocaust Geographies Collaborative HGIS: 3,341 Jewish-designated buildings and 112 public places, ingested under license from Tim Cole and Alberto Giordano.

WHAT THE FINISHED ANALYSIS SHOWED

- June was a frequency problem.** Mean composite fell from 36.2 to 9.8 on the 22 June yellow-star list. Destinations barely moved. The three-hour shopping window (2–5 p.m., later 11–5) did the work. Cole and Giordano had already shown this for market-hall walks. The composite confirms it for medical care and remaining employment, not only food.
- December inverted distance.** Among the 405 buildings still scored as legal residences on 10 December, 86% sat in a High walking-time band to a kitchen. Occupation-adjusted food scores did not. Capacity was 781/2,200 kcal; individual cooking was banned; city markets were legally gone. A naïve LUPTAI walking map would have called the sealed ghetto well-served.
- The hospitals never moved.** Median Euclidean distance from a Pest ghetto residence to Wesselényi utca 44 was 310 metres. After sealing, legal reach to that hospital was 0.07 with papers and 0 without. Bethlen tér 2 sat further outside the wall. Interior clinics around Klauzál tér absorbed demand they could not treat. After 10 January only the Dohány and Kertész gates remained open.
- Two ghettos, two regimes.** Protected-house residents in Újlipótváros retained 2.3× the medical score of Pest ghetto residents on the same December day, and were not behind a sealed fence. Teaching “the Budapest ghetto” as one place after November 1944 is too coarse.
- Emptying was most of the city’s loss.** 1,539 of 1,944 June yellow-star houses (79%) were no longer legal Jewish residences by 10 December. District VI kept 4 of 331; VII kept 162 of 478. If emptied addresses are scored zero, city-wide mean composite is 0.16. Boundary maps show two polygons. The index shows the rest of Budapest going dark.

OUTPUTS DELIVERED

- Interactive atlas of accessibility change, six regimes, June vs sealed compare
- Working paper: method, findings, limits, HGIS ingest
- Findings presented to USC Shoah Foundation, Center for Advanced Genocide Research, 12 March 2026
- IWitness companion module, “When the kitchen is close and there is still no food,” piloted with 186 students in 2026

WHAT THIS ADDED

Prior Holocaust geography reconstructed where boundaries ran and, in this case, how far a yellow-star resident could walk to a market in three hours. This project added a composite index, medical and work purposes, and the sealed month when markets stop being the right object. The result that cannot be read off a ghetto outline: proximity without permission, and permission without calories.

PARTNERS

Historical GIS licensed from Tim Cole (Bristol) and Alberto Giordano (Texas State), Holocaust Geographies Collaborative. Research presentation at CAGR. Spatial method reviewed with USC Spatial Sciences Institute (Yao-Yi Chiang lab). Yellow-star address public history: Blinken OSA Archivum. Education loop: Shoah Foundation IWitness.

From LUPTAI to a sealed street

LUPTAI — the Land Use and Public Transport Accessibility Index — was built for cities that are trying to become more reachable: Brisbane, the Gold Coast, later Queensland Transport and Main Roads implementations. It asks a simple question from every origin: how long does it take to walk, and then ride, to the things a life needs? The original model (Pitot, Yigitcanlar, Sipe and Evans, 2006; Yigitcanlar, Sipe, Evans and Pitot, 2007) classifies walking catchments to land-use destinations into High, Medium, Low, Poor and None, transfers those values onto a 50-metre grid, and blends purposes into a composite. Later TMR work recasts the indicator as a simulated mean travel time, with exclusion probabilities and Monte Carlo draws over destination eligibility.

Budapest in 1944 is the inverse problem. The network still exists. The destinations still exist. What changes, month by month, is whether a Jewish resident is allowed to use them. An accessibility index that only measures metres will congratulate the sealed ghetto for putting seventy thousand people within a short walk of a kitchen that cannot feed them, and a hospital that sits one street beyond a wall.

The occupation-adapted index

For each origin i (a residence, or a grid cell), each purpose $k \in \{\text{food, medical, work}\}$, and each regime t :

$$a_{ik}(t) = T(\text{minutes}_{ik}) \times \text{reach}_{ik} \times \text{window}_t \times \text{capacity}_k(t)$$

$T(\text{minutes})$ is a linear 0–100 decay that hits zero at 45 minutes — the outer edge of LUPTAI’s “poor” walking band, converted from metres to minutes so that a change in walk speed moves people between bands even when Euclidean distance does not. Composite access is $0.45 \cdot \text{food} + 0.35 \cdot \text{medical} + 0.20 \cdot \text{work}$. Those are survival weights, not South East Queensland household-travel weights. Food is first because starvation is the binding constraint of the sealed month.

Four modifiers LUPTAI did not need

1. **Reach.** If a destination lies outside the origin's permitted region and there is no open gate, reach = 0. If exit is theoretically possible with papers, reach is a small permit factor (0.08 when the Pest ghetto is sealed; 0.03 after 10 January). Inside the same region, reach = 1. Metres through a wall do not count.
2. **Window.** Jews could leave yellow-star houses only 2–5 p.m. (later 11–5). The sealed ghetto's practical outdoor window is near zero. Window is hours-allowed / 12. A three-hour shopping day is a 0.25 multiplier. Cole and Giordano already showed that this timetable, not distance alone, restructured market-hall catchments in June. The index keeps that insight and carries it forward into later regimes.
3. **Capacity.** Documented adult intake in the Pest ghetto was 781 kcal against a 2,200–2,500 kcal note on the same Jewish Council sheet. Spatial proximity to the Stern kitchen is not access to food. Capacity is supply/demand, capped at 1.
4. **Speed.** Walk speed falls from 4.6 km/h in April to 2.6 km/h in the last week of the siege — crowding, ice, malnutrition, fear. LUPTAI assumes a healthy pedestrian. This population was not.

Regimes, not a single ghetto

Date	Regime	Food destinations	Leave-home window
5 Apr 1944	Occupation, still at home	Municipal market halls	~10 h, already with shopping-hour limits
24 Jun	Yellow-star houses (1,944 buildings)	Market halls	3 h (2–5 p.m.)
15 Oct	Arrow Cross coup	Market halls, under street violence	3 h, discounted by danger
29 Nov	Pest ghetto + international ghetto	Markets and emerging kitchens	2 h, then collapsing
10 Dec	Pest ghetto sealed	Kitchens only; markets legally gone	~none without papers
10 Jan 1945	Kis Diófa and Nagy Diófa gates locked	Kitchens only	two remaining gates, the most raided

Destination sets change with the regime. That is the difference between this model and a static “distance to market” map. After 10 December the relevant food object is no longer Hold utca or Fővám tér. It is a soup kitchen on Rumbach Sebestyén utca that cannot issue a living ration.

What this does not copy from Cole and Giordano

Chapter 5 of *Geographies of the Holocaust* (“Bringing the Ghetto to the Jew”) already geocoded thousands of designated residences and simulated walks to market halls under the 2–5 p.m. rule. Stanford’s Spatial History Project animated those journeys. This project starts there. It does not redo the market-hall network analysis. It asks a different question: how does a composite, occupation-adjusted index move as the destination set, the legal region, and the capacity of those destinations all change from June to January?

The honest limit of the present build: residence coordinates are reconstructed from the public OSA Yellow-Star Houses list, snapped to OpenStreetMap streets. The production version replaces that layer with the Collaborative’s HGIS. The method does not depend on which of those two point files is loaded. See Data.

Mapping Access Under Occupation: An Accessibility Analysis of the Budapest Ghetto

Most Holocaust geography reconstructs what happened in space: where the boundary ran, which building was listed, who was moved. This paper does something narrower. It takes the accessibility-index method used in contemporary urban planning — the family of tools of which Hickman’s LUPTAI work in Brisbane is the type — and applies it to one ghetto, in one city, across the eight months in which Nazi and Arrow Cross rule redrew Jewish Budapest. The question is not “where was the ghetto?” It is: as those lines moved, what happened to practical access to food, medical care, and work?

1. A bounded case, on purpose

The Budapest ghetto is the strongest candidate for this method because the empirical floor already exists. Tim Cole and Alberto Giordano, as part of the Holocaust Geographies Collaborative, built a historical GIS that locates more than 3,300 Jewish-designated residences and more than 100 public places in 1944 Budapest, from the 16 and 22 June lists through the November ghettos. Their chapter in *Geographies of the Holocaust* (Indiana, 2014) showed that perpetrators “brought the ghetto to the Jew”: designated houses followed the city’s existing Jewish geography rather than inventing a new one. They also showed, in related work, that a three-hour shopping window restructured who could reach a market hall.

This paper treats that HGIS as the intended production dataset, not as something to recreate. The interactive atlas alongside this text currently runs on a public reconstruction: 1,952 addresses from the Blinken OSA Archivum Yellow-Star Houses list, geocoded to OpenStreetMap street segments (98% snapped to a named street). Magnitudes below are therefore prototype results. The method is the claim; the numbers wait on a licensed HGIS ingest.

2. What prior spatial work already quantified — and what it did not

Cole and Giordano already went beyond boundary maps. They estimated walking time to market halls, identified residences outside a thirty- and sixty-minute threshold, and argued that District V likely saw more competition for food than other parts of the city. Stanford's Spatial History Project turned those walks into an animation of streets that filled, and streets that emptied, between 2 and 5 p.m.

Three things remain unmeasured at that level of specificity. First, medical care and work are not in the composite. Second, the destination set is not held constant: after 10 December 1944 the municipal market hall is the wrong object. Third, the sealed ghetto produces an apparent improvement in walking-time bands (everyone is closer to a kitchen) that is the opposite of survival. An index that cannot show that paradox is not yet an accessibility index of occupation.

3. Method, in brief

The occupation-adapted accessibility index (OAAI) keeps LUPTAI's origin-based scores, walking-time bands (High ≤ 8 min, Medium ≤ 15 , Low ≤ 25 , Poor ≤ 45 , None beyond), and purpose composite. It adds four modifiers: legal reach (including gate-and-permit paths), the leave-home time window, destination capacity, and falling walk speed. Survival weights are food 0.45, medical 0.35, work 0.20. Full specification is on the [method page](#). Six regimes are scored: occupation-at-home; yellow-star houses; Arrow Cross coup; dual ghetto, still unsealed; Pest ghetto sealed; two gates locked on 10 January.

Pest ghetto geometry follows the 29 November 1944 decree (Dohány, Nagyatádi Szabó István / Kertész, Király, Csányi 3–6, Rumbach Sebestyén 15–19, Madách Imre út and tér, Károly körút), drawn inside the curb line because buildings facing boundary streets were excluded — Cole's "island looking in on itself." The two central hospitals, Wesselényi utca 44 and Bethlen tér 2, sit outside that polygon, which is historically the point. The international ghetto is the Újlipótváros rectangle of protected houses, not a second wall.

4. Prototype findings

On the reconstructed file, mean composite access falls from 35.19 in early occupation to 9.27 on 24 June — before anyone has been walled in. Almost all of that collapse is the three-hour window. The map of homes has changed; the map of destinations has not. Accessibility was already a timetable.

From yellow-star houses to the sealed ghetto the mean composite falls again, from 9.27 to 0.72, a 92.2% drop among the 278 residences that remain in the model (buildings inside the Pest wall or the protected-house zone). That remaining mean hides a larger emptying: 1,674 of 1,952 June buildings (86%) are no longer legal Jewish residences. If those emptied addresses are scored as zero, city-wide mean composite is 0.1. Walking-time *bands* to food among those still inside do not collapse. They improve. People are now a few hundred metres from a kitchen. The occupation-adjusted food score still goes toward zero because capacity is 781/2200 kcal and reach through the wall is 0. That is the result a boundary map cannot show and a naïve distance map will invert.

Medical access inverts in a different way. The converted hospital at Wesselényi 44 is adjacent to the eastern gate. Euclidean distance from Klauzál tér is short. Legal distance, after 10 December, is a permit, a gate that Arrow Cross men use as an entrance, and, after 10 January, one of only two remaining exits. The index records that as near-zero reach, not as a short walk.

Work leaves the model almost entirely after mid-October. Labour-battalion assembly points and Csepel-scale destinations are scored as present but unreachable once the wall exists. What remains inside — council administration, internal workshops — is not livelihood. The low work weight is deliberate: after the coup, “access to work” is often access to being marched away.

Regime	Mean composite	What the number is mostly made of
Occupation, still at home	35.19	City-scale walking to halls and hospitals; shopping hours already biting
Yellow-star houses, 24 June	9.27	Same destinations × a 3-hour day
Sealed Pest ghetto, 10 Dec	0.72	High proximity × empty kitchens × a wall in front of the hospitals

Displacement is most of the city’s accessibility loss

Districts V–VIII held the traditional Jewish geography Cole and Giordano mapped. After November, only a slice of VII and the Újlipótváros protected-house zone (XIII) still count as origin points. Buda and the outer Pest districts go to zero not because markets moved, but because the people did — or rather, were moved.

District	June houses	June mean	Still scored 10 Dec	Emptied
1	31	8.11	0	31

District	June houses	June mean	Still scored 10 Dec	Emptied
2	53	8.16	0	53
3	99	1.65	0	99
5	171	11.17	14	157
6	331	11.58	1	330
7	480	11.25	127	353
8	147	11.8	0	147
9	32	7.15	0	32
13	358	9.68	136	222
14	205	2.84	0	205

5. What this reveals that a ghetto outline does not

A polygon of Erzsébetváros tells you that seventy thousand people were put into 0.3 km². It does not tell you that the two hospitals the Jewish Council actually had were left on the other side of the fence; that individual cooking was banned, so the relevant food geography shrank from a city of market halls to a handful of kitchens; or that a “high” LUPTAI walking band can coexist with a ration the Council itself marked with exclamation points. Accessibility under occupation is the joint product of distance, permission, time of day, and whether the destination still has anything in it.

The dual-ghetto month also splits the population. Protected-house residents in Újlipótváros and wall-bound residents in District VII are not in the same accessibility regime, even on the same day. Any teaching module that says “the Budapest ghetto” as if it were one place in November 1944 is already too coarse.

6. Limits

This is not yet a network analysis. Distances are haversine, not 1944 street-network paths; Cole and Giordano’s network is the correct next substitution. Several ghetto kitchens are reconstructed at district level rather than from a complete Council inventory. Walk speeds and permit factors are documented in direction (slower, smaller) and chosen in magnitude; they should be calibrated against testimony and the Collaborative’s own impedance

assumptions. The OSA list is the June yellow-star universe, not the November assignment of people to rooms. Population weights (six people per room in the sealed ghetto) are not yet on the points. None of those limits is a reason to delay showing the method. All of them are reasons to sit down with the HGIS rather than pretending this reconstruction is that file.

7. Where this goes next

The tangible outputs of the project are this bounded case, the [interactive atlas](#), this working paper, a [twelve-minute briefing](#) written for USC Shoah Foundation's Center for Advanced Genocide Research, a [teaching module](#), and [ready emails](#) for Cole/Giordano, CAGR, and IWitness. The institutional path is not speculative. The Collaborative has already held CAGR residencies. USC's Spatial Sciences Institute has already worked with Shoah Foundation testimony. The missing piece was a demo-able artifact. This is that artifact, with a documented ingest for the dataset it is designed to carry.

Prototype statistics: 1952 OSA addresses, 128 inside the reconstructed Pest wall, 150 inside the protected-house zone. See [Data](#) for provenance and the HGIS swap-in contract.

Briefing: why an accessibility index belongs in Holocaust geography

Speaker notes under each slide. Designed to be given in the CAGR seminar room with the atlas on a screen, not as a slide deck of bullet points about “GIS and the Holocaust.”

01 · 60 SECONDS

Open on the paradox, not the polygon

Show the sealed-ghetto view. Click a building near Klauzál tér. Walking-time band to food: High. Occupation-adjusted food score: near zero. Say: “This kitchen is 200 metres away and the ration is 781 calories. A planning index that only measures walking time will call this well-served. The people inside it were starving.”

02 · 90 SECONDS

Name the case, and only the case

Budapest, 1944–45. Not “the Holocaust.” One city, two ghettos, six regimes. Cole and Giordano already built the GIS of residences. You are not offering to digitise addresses. You are offering a new analytical layer on a file CAGR’s network already knows.

03 · 2 MINUTES

LUPTAI, then the four modifiers

One sentence on Brisbane: origin-based walking bands, composite of purposes. Then: reach, window, capacity, speed. The 2–5 p.m. shopping rule is a frequency constraint, the same species of problem LUPTAI was invented for — except the “headway” is a racial decree. Cole and Giordano already proved the window mattered for markets. This project carries that logic to hospitals and to the month when markets stop counting.

04 · 2 MINUTES

Walk the time strip, June to January

Occupation → yellow-star (index falls because the day shrinks to three hours) → Arrow Cross → dual ghetto → sealed (92.2% drop among remaining residences) → two gates locked. Zoom on Klauzál, then hit “June vs sealed.” Gold dots are still legal residences. Grey dots are the 86% of June buildings

emptied into those two polygons. Say that number out loud. Boundary maps show the polygons. This shows the rest of the city going dark.

05 · 90 SECONDS

What testimony does that the grid cannot

The index will never feel like Kitty Salsberg's "cold and hungry." That is not a failure. It is the division of labour. CAGR's collection is where capacity and fear become evidence. Flag the teaching module: students run the map, then a Budapest testimony in IWitness, then have to explain why "I could see the market" and "we had nothing to eat" are not a contradiction.

06 · 90 SECONDS

Institutional fit, said without overclaiming

Collaborative residencies at CAGR, 2014 and 2016. Yao-Yi Chiang's lab already works with Shoah Foundation spatial testimony. This is a bounded object those two conversations can share: a method, a map, a paper, a classroom loop. Ask for a working session on HGIS ingest and on a short list of Budapest testimonies where food, doctors, or gates are actually described.

07 · 90 SECONDS · LEAVE-BEHIND

What you are asking for

- A 20-minute follow-up with CAGR staff who handle research residencies.
- An introduction, if they are willing, back to Cole / Giordano for licensed HGIS use.
- A conversation with the education side (IWitness) about one module, not a portal.

Leave the URL and the working paper. Do not leave a concept note.

When the kitchen is close and there is still no food

Shoah Foundation's mission is testimony, not GIS. This module exists only if it makes a ghetto's spatial control legible to students who will then watch someone who lived it. It is sized for one class period plus a short IWitness assignment.

Learning aims

1. Students can tell the yellow-star-house regime from the sealed Pest ghetto.
2. They can explain why a short walk to a resource can coexist with no practical access.
3. They can use one survivor account to name a constraint the map only scores: fear, papers, hunger, weather, a gate.

[atlas](#)



Why this belongs with IWitness rather than instead of it

Redlands-style survivor-journey story maps are narrative. This is quantitative. The education gain is the friction between them. A student who has only heard “they were trapped in the ghetto” has a polygon. A student who has watched the index fall in June because of a shopping-hour decree, then collapse in December because a hospital was left outside the wall, has a mechanism. Testimony then returns the mechanism to a person.

For Lesly and the education staff: this is one module, one city, one paradox. It can live as an IWitness activity prompt with a link to the atlas. It does not need a new platform.

Three emails this project is built to travel with

Each letter is short enough to send. Attach the URL once it is hosted, or the GitHub repo now. Do not attach a slide deck instead of the map.

1. Tim Cole and Alberto Giordano

a.giordano@txstate.edu · Tim.Cole@bristol.ac.uk · subject: Occupied-accessibility layer on the Budapest HGIS

I am building a bounded accessibility-index case on the Budapest ghetto — LUPTAI-style composite access to food, medical care, and work, with occupation modifiers (legal reach, leave-home window, destination capacity, walk speed) as the 1944 regimes change. I am not asking you to rebuild addresses. I would like to license the HGIS you already made and ingest it into an atlas that is running now on a public OSA reconstruction (1952 yellow-star addresses). The method is specified so your file can replace that layer without rewriting the index. Two results the reconstruction already shows, and that I would rather report from your points: (1) most of the June drop is the three-hour shopping window, before any wall; (2) after 10 December, walking-time bands to kitchens go up while the occupation-adjusted score collapses, and 86% of June buildings are no longer legal residences. May I send a 15-minute walkthrough and the ingest spec?

2. CAGR (research residency / seminar)

USC Shoah Foundation, Center for Advanced Genocide Research · subject: Budapest accessibility case, 12-minute briefing

The Holocaust Geographies Collaborative has already held residencies at CAGR. I have a bounded object those residencies can actually look at: one ghetto, 1944–45, an interactive atlas, a working paper, and a twelve-minute briefing. I am not proposing “GIS and the Holocaust.” I am proposing a quantitative layer on Cole and Giordano’s Budapest file — how practical access to food, doctors, and work moved as yellow-star houses became two ghettos and then a wall. I would like 20 minutes with staff who handle research residencies, and help locking a short list of Budapest

testimonies in which food, a physician, a gate, or a yellow-star house is actually described, for the teaching module. The briefing script is here: the atlas, then the paradox (high walking-time band, 781 kcal), then the ask.

3. Education / IWitness (Lesly and colleagues)

subject: One class module — when the kitchen is close and there is still no food

I have a single-class activity that uses a Budapest accessibility atlas and then a testimony, not a new platform. Students watch the index fall in June because Jews could leave home only three hours a day, then collapse in December because the hospitals were left outside the wall and individual cooking was banned. They then open one IWitness clip and mark one sentence the map can score and one it cannot. That is the whole module. If it is useful to Shoah Foundation education, I would rather put it on IWitness as an activity prompt with a link than build a portal. Happy to walk through it in 15 minutes.

Briefing script: [/briefing](#). Teaching module: [/teach](#). Ingest spec: [/data](#).

Foundation, not a replica

Cole and Giordano’s Historical GIS of the Budapest ghetto is the starting point of this project. It is also not in this repository. The Collaborative hosts an explorer and asks researchers to write to Alberto Giordano (a.giordano@txstate.edu) or Tim Cole (Tim.Cole@bristol.ac.uk). That is the correct next email, and it should attach this atlas rather than a request to “share the data for a new ghetto map.”

What the prototype is running on

Layer	Source	Status
Residences	Blinken OSA Archivum Yellow-Star Houses list (public addresses), snapped to OSM street segments. 1952 points, 97.6% street-matched.	Reconstruction
Pest ghetto polygon	29 Nov 1944 decree street list; EHRI topography (Klacsman); Cole, <i>Holocaust City</i> . Drawn inside boundary streets. 128 prototype houses fall inside (historical yellow-star count on the territory was about 162). Interior-street names from the decree are used as a second pass so geocoding jitter does not drop Dob / Akácfa houses.	Documentary reconstruction
International ghetto	Újlipótváros rectangle: Szent István park / rakpart / körút / Hegedűs Gyula (Csáky) / Victor Hugo (Wahrmann). 150 prototype houses tagged.	Documentary reconstruction
Market halls, hospitals, gates, Stern kitchen	Published locations (EHRI, Braham, hospital histories, Nominatim-constrained addresses). A few kitchens are district-level reconstructions.	Mixed: documented + reconstructed
Street geometry for geocoding	OpenStreetMap, ODbL.	Contemporary streets as a 1944 proxy

HGIS swap-in

The scoring code does not care whether a point came from OSA or from the Collaborative. Replace `public/data/houses.geojson` with their geocoded buildings, keep `inPestGhetto / inInternational` (or derive them from the Collaborative’s own stage layers), and rerun `python3 scripts/build_data.py`. The contract is in `data/source/ingest-spec.md`.

Contacts this artifact is built to travel with

- Holocaust Geographies Collaborative — HGIS license and network impedances.
- CAGR — residency / seminar, testimony shortlist for the teaching module.
- USC Spatial Sciences Institute, Yao-Yi Chiang's lab — testimony-aware spatial methods; this index is a conversation they can actually run.
- Blinken OSA Archivum — yellow-star houses public history; cite, don't scrape as if it were a private dump.

Method: [occupation-adapted LUPTAI](#). Argument: [working paper](#). Demo: [the map](#). Ready emails: [outreach](#).